Lesson 7

AIR QUALITY AND SUPERVISION

INTRODUCTION

- **Air quality** refers to the condition of the air in a particular place based on the **amount and type of pollutants present**.
  - **Good air qual**ity supports human health and ecosystems,
  - **Poor air quality** can cause respiratory diseases, environmental damage, and climate-related problems.
- This lesson discusses the **Earth's atmosphere, air pollution, air quality management, and methods of controlling air pollution**.

LEARNING OBJECTIVES

At the end of the lesson, students should be able to:

1. **Describe** the structure and composition of the Earth's atmosphere.
2. **Identify and explain** major sources, types, and effects of air pollution.
3. **Explain** the principles and practices of air quality management.
4. **Evaluate** appropriate methods for preventing and controlling air pollution.

THE EARTH'S ATMOSPHERE

The **atmosphere** is the **layer of gases** surrounding Earth. It provides the conditions necessary for life.



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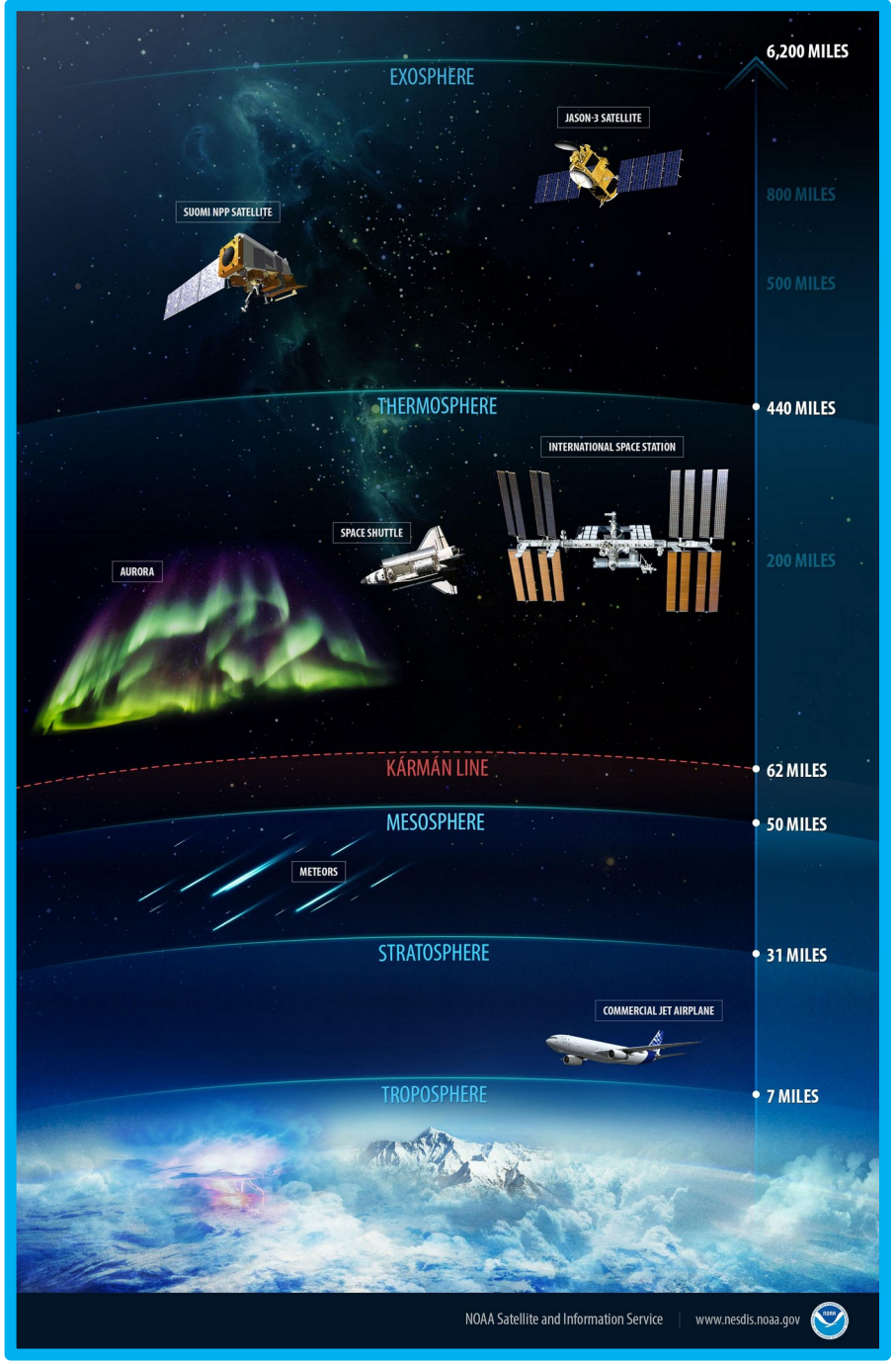
COMPOSITION OF THE ATMOSPHERE

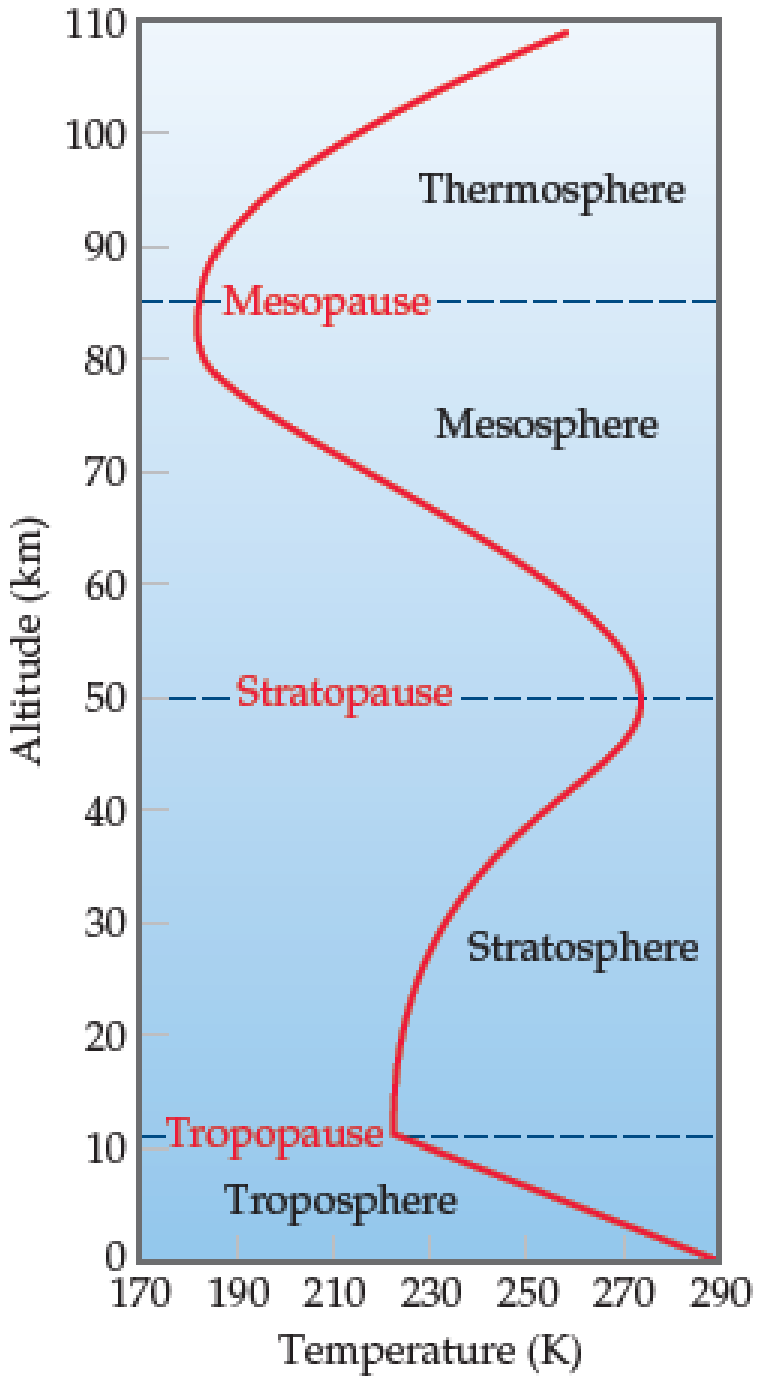
Gas	Approximate Proportion	Importance
Nitrogen (N_2)	**78%**	Important in the **nitrogen cycle** and production of proteins.
Oxygen (O_2)	**21%**	Required for **respiration** and **combustion**.
Argon (Ar)	**0.93%**	Inert gas with limited biological activity.
Carbon dioxide (CO_2)	**~0.04%**	Used by plants in **photosynthesis**; **Greenhouse gas**.
Other gases	**Small amounts**	Include neon, helium, methane, ozone, and others.
Water vapor	**Variable**	Important in **weather, clouds**, and the **water cycle.**

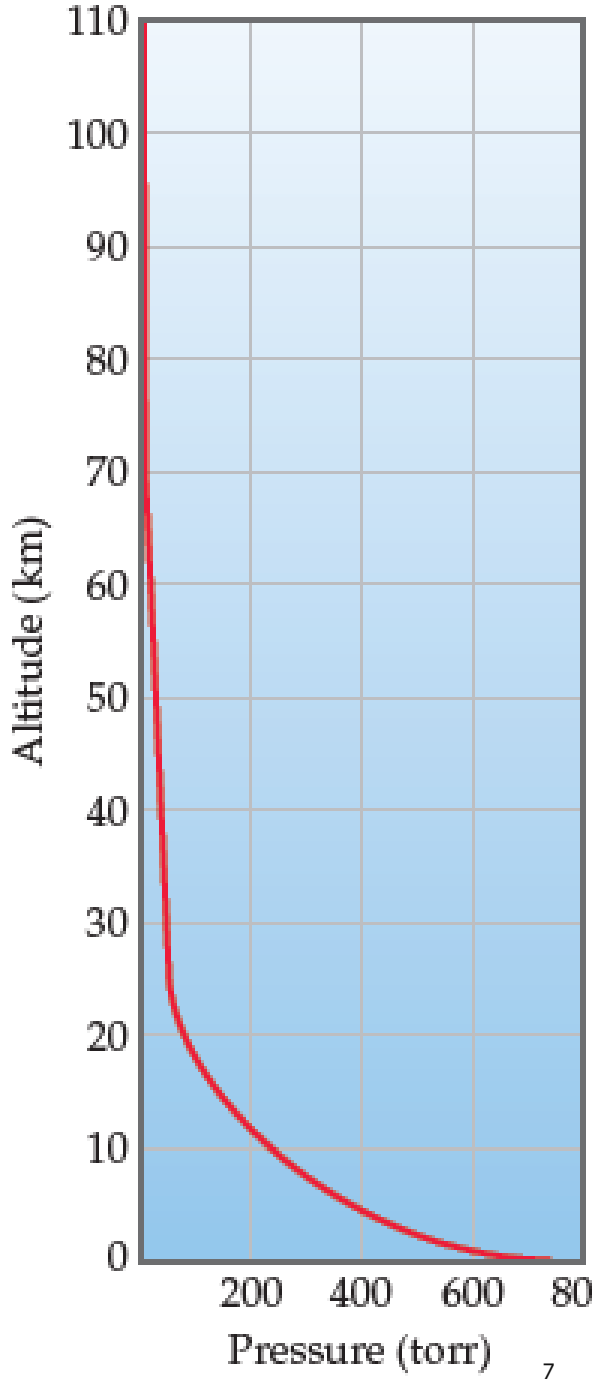
LAYERS OF THE EARTH'S ATMOSPHERE

- The atmosphere is divided into several layers according to **changes in temperature** and **other characteristics.**

Layer	Temperature trend	Important Characteristics
Troposphere	↓ Decreases	**Weather occurs here.**
Stratosphere	↑ Increases	**Contains the ozone layer.**
Mesosphere	↓ Decreases	Many meteoroids burn up here.
Thermosphere	↑ Increases	Very high temperatures; auroras occur here
Exosphere	Gradually transitions	Outermost layer; gradually transitions into space

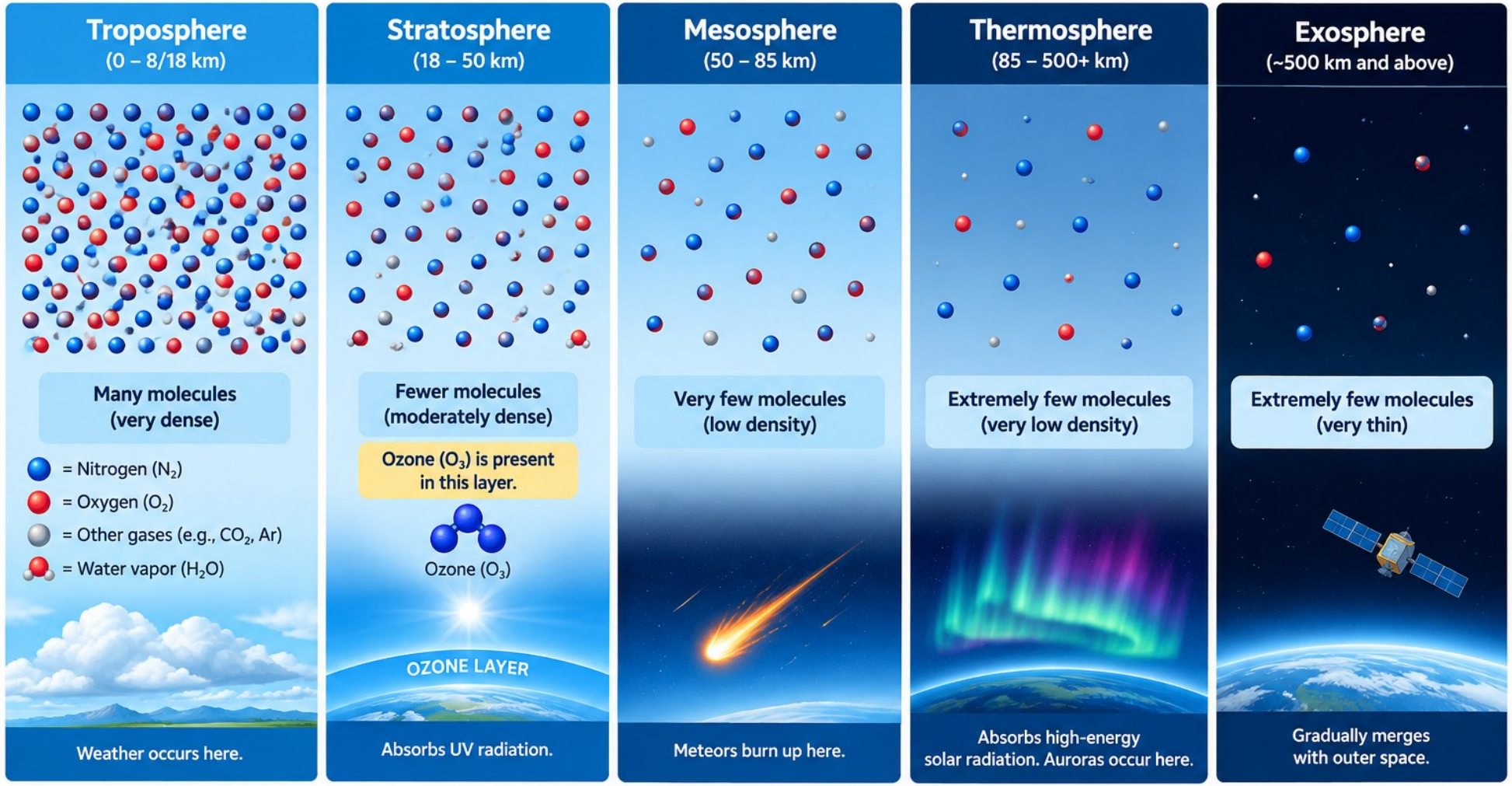

6,200 MILES
EXOSPHERE
JASON-3 SATELLITE
SUOMI NPP SATELLITE
800 MILES
500 MILES
THERMOSPHERE
440 MILES
INTERNATIONAL SPACE STATION
SPACE SHUTTLE
AURORA
200 MILES
KÁRMÁN LINE
62 MILES
MESOSPHERE
50 MILES
METEORS
STRATOSPHERE
31 MILES
COMMERCIAL JET AIRPLANE
TROPOSPHERE
7 MILES
NOAA Satellite and Information Service
www.nesdis.noaa.gov


Thermosphere
Mesopause
Mesosphere
Stratopause
Stratosphere
Tropopause
Troposphere
Altitude (km)
Temperature (K)
110
100
90
80
70
60
50
40
30
20
10
0
170
190
210
230
250
270
290


Altitude (km)
Pressure (torr)
110
100
90
80
70
60
50
40
30
20
10
0
200
400
600
80

Molecules in Each Layer of the Atmosphere

The number of air molecules decreases as we go higher. The diagrams below show a relative comparison of molecular density in each layer (not to scale).



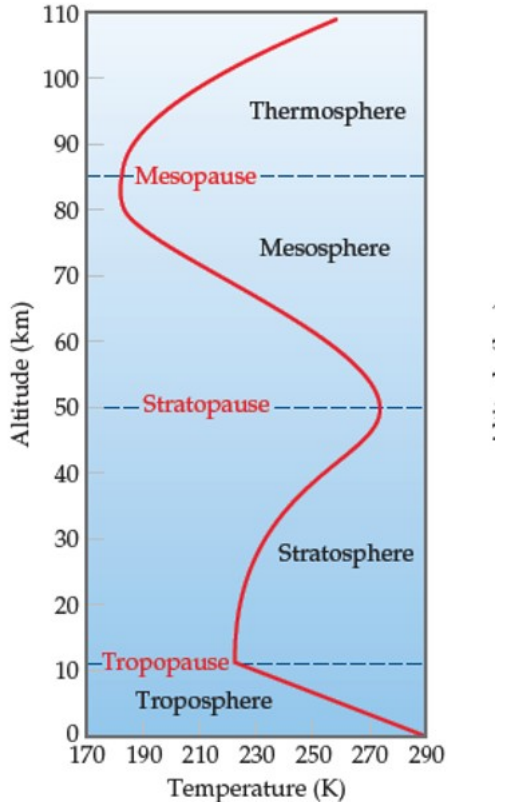
= Nitrogen (N_2) = Oxygen (O_2) = Argon (Ar) = Carbon dioxide (CO_2) = Water vapor (H_2O)

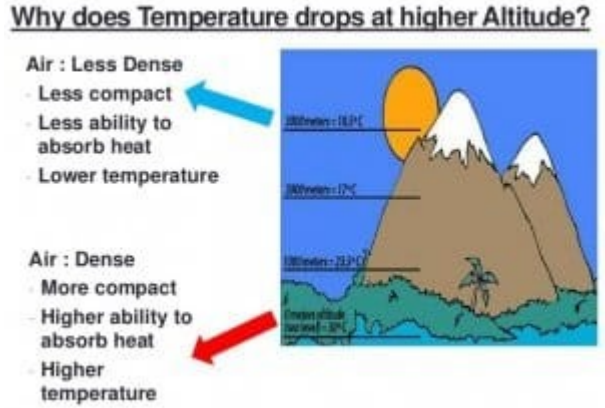
1. TROPOSPHERE

- The **troposphere** is the lowest atmospheric layer and the layer in which people live.
- It contains most of the atmosphere's water vapor and is where:
  - clouds form
  - rain occurs
  - winds develop and
  - most weather takes place.
- **Air pollution that directly affects human health is primarily concentrated in the troposphere.**

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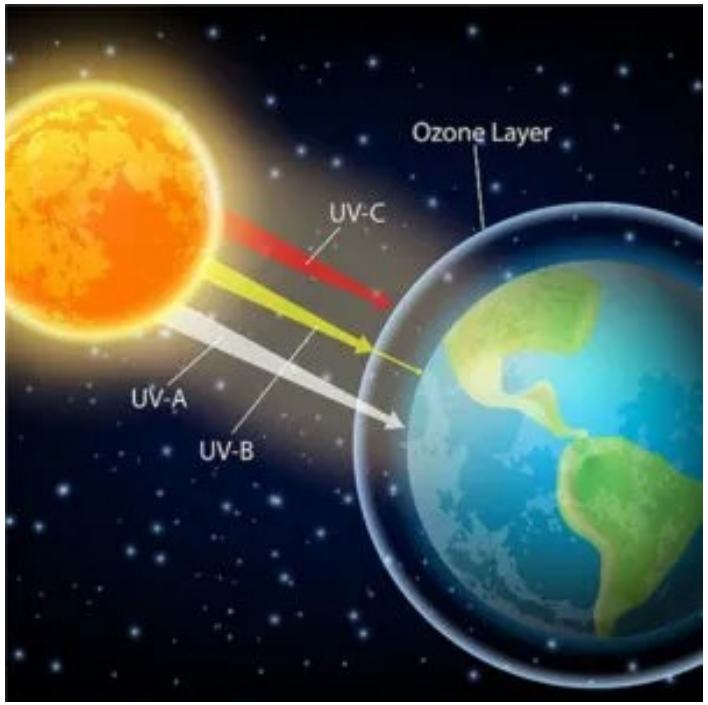
- As **altitude increases**, **temperature** generally **decreases.**
- This happens because the Earth's surface is the main source of heat for the lower atmosphere. Air farther from the surface receives less heat.
- **Example: Mountain areas** are **generally cooler** than **areas near sea level.**

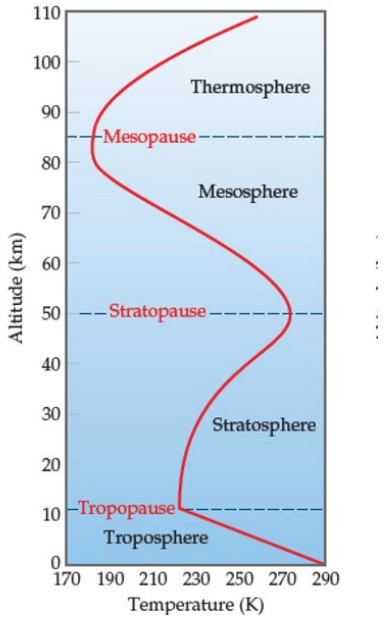




2. STRATOSPHERE

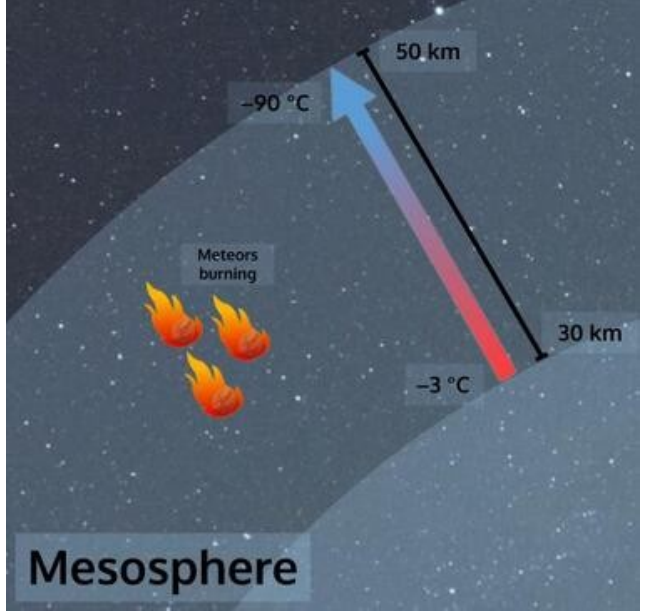
- The **stratosphere** lies above the troposphere. It contains the **ozone layer**, which **absorbs** much of the Sun's harmful ultraviolet **(UV) radiation.**
- Temperature **increases as altitude increases**.
- The major reason is the presence of **ozone (O_3)**, which absorbs ultraviolet (UV) radiation from the Sun and converts some of that energy into heat.
- **UV radiation → absorbed by ozone → heat → temperature increase**

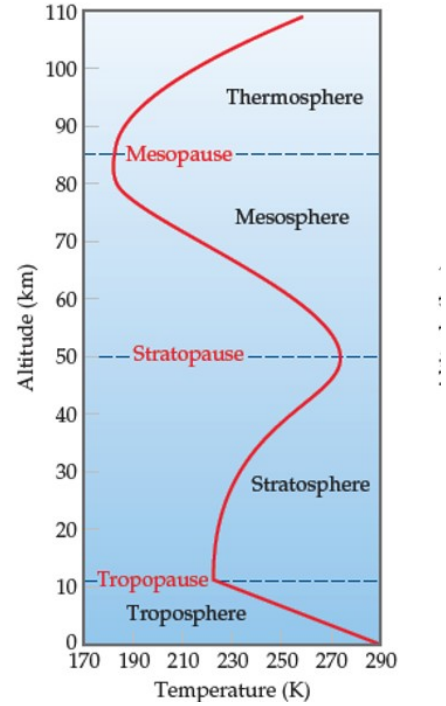




3. MESOSPHERE

- The **mesosphere** is the layer where many **meteoroids** entering Earth's atmosphere **burn (**producing what we commonly call **shooting stars**) because of friction with atmospheric gases.
- Temperature **decreases as altitude increases**. The mesosphere is one of the coldest regions of Earth's atmosphere.
- **Mesosphere = little ozone → less absorption → cools upward**





WHY IS THE MESOSPHERE COLD EVEN THOUGH IT IS CLOSER TO THE SUN?

- Atmospheric temperature depends more on **how much solar energy is absorbed** than simply on distance from the Sun.
  - The air becomes **much thinner**.
  - **There is less ozone available to absorb UV radiation. Less solar energy is converted into heat.**

What about meteors?

- The fact that meteors burn in the mesosphere does **not** mean the entire mesosphere is hot. The meteor is heated mainly because of its extremely high-speed interaction with atmospheric gases.

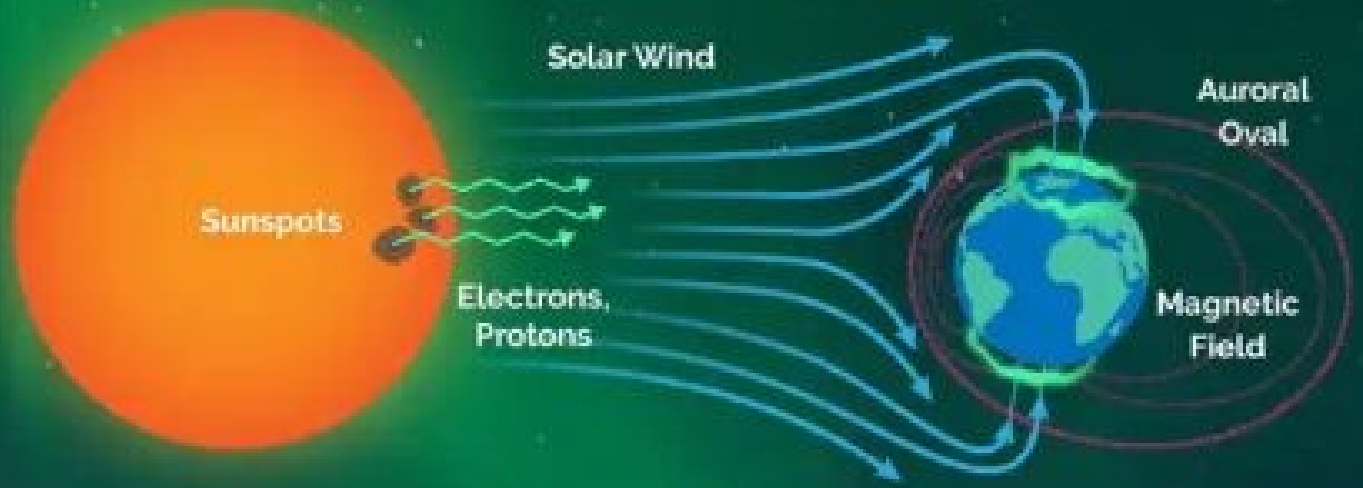
4. THERMOSPHERE

- **Auroras** occur in this region.
  - Borealis – North
  - Australis - South
- Collision with oxygen
  - Green and red emission
- Collision with Nitrogen
  - Low level red and very high blue/violet



HOW AURORAS FORM

Auroras are the result of the interaction of space weather with Earth's magnetic fields



At any given moment, the sun is ejecting charged particles from its upper atmosphere, creating what's called the solar wind.

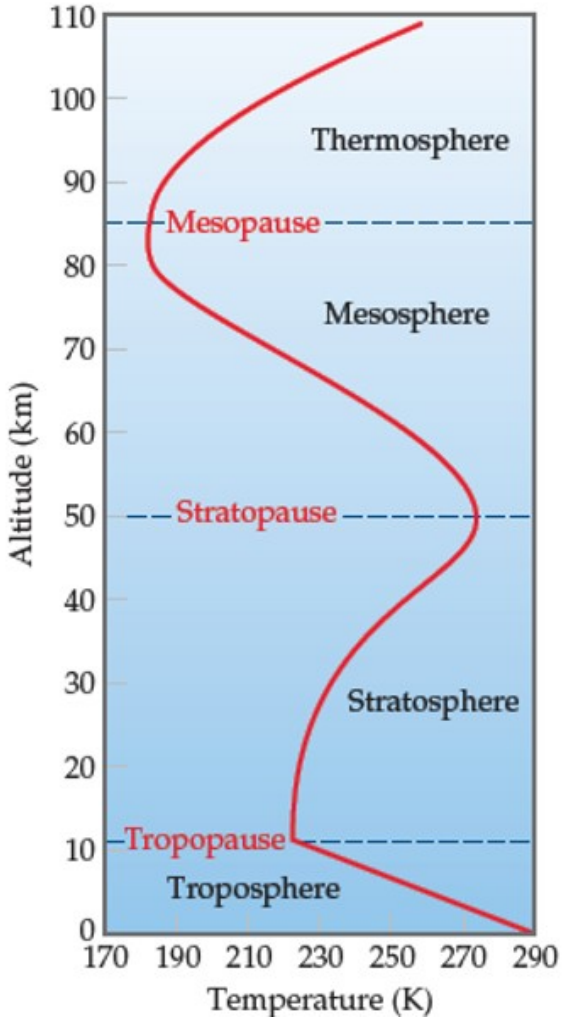
These charged particles travel through space and some are captured by the Earth's magnetic field.

These particles enter Earth's atmosphere at the poles, collide with molecules present in the atmosphere, and emit light

Cont.

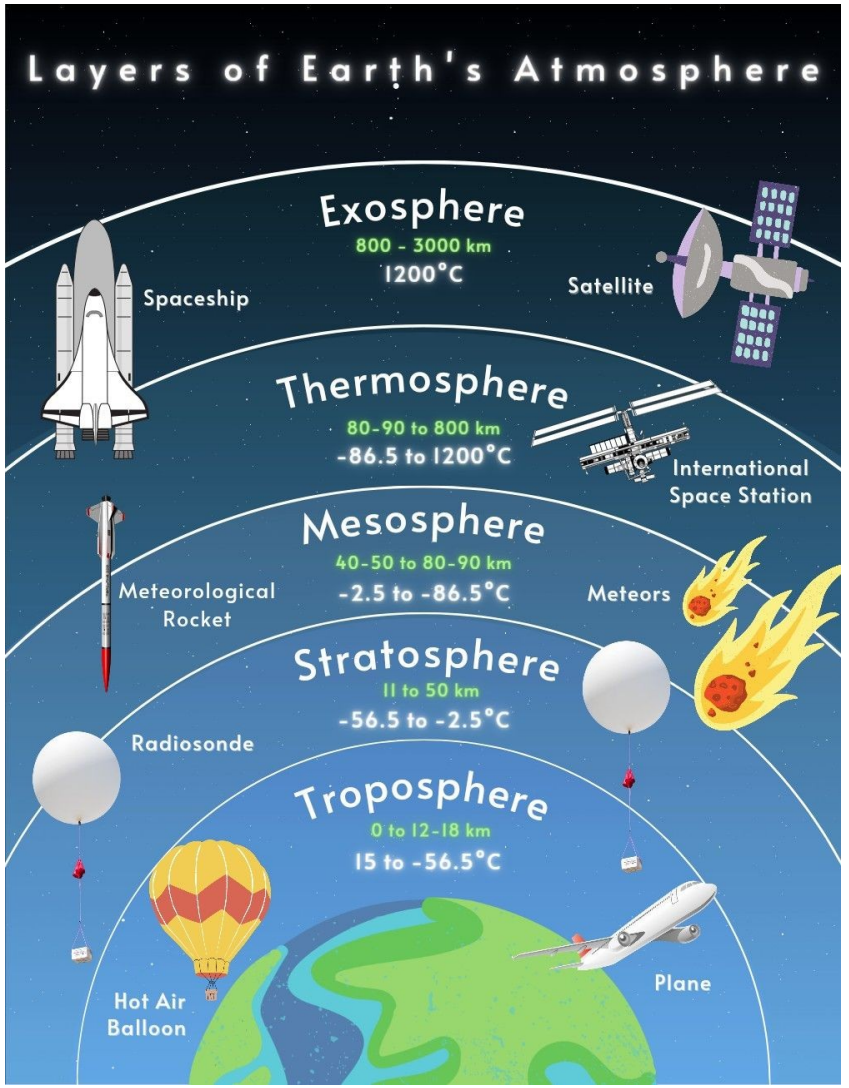
The **thermosphere** contains **very thin air** and can **reach extremely high temperatures**.

- Temperature **increases as altitude increases**.
  - The temperature of the thermosphere increases with altitude because the gases in this layer absorb high-energy solar radiation, particularly extreme ultraviolet radiation and X-rays. This energy makes the gas particles move very rapidly, resulting in very high temperatures.
  - However, the thermosphere is extremely thin, so there are very few particles to transfer heat. Therefore, although the measured temperature is very high, the thermosphere would not feel like a very hot atmosphere on Earth.



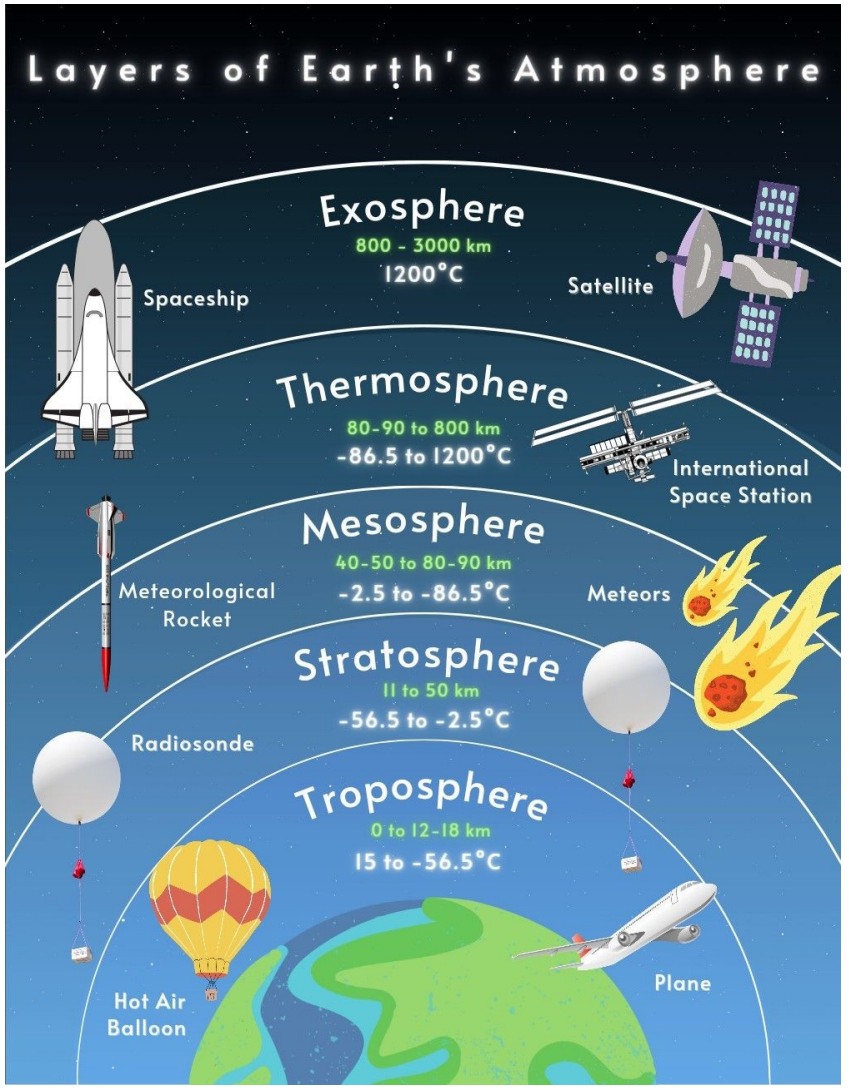
Cont.

- Some **spacecraft and satellites** orbit within or pass through this region.
- Many satellites, especially those in Low Earth Orbit, are in the thermosphere or above it. The ISS, for example, orbits within the thermosphere at about 400 km altitude. Higher-orbit satellites are far above the main layers of Earth's atmosphere.
- Although the thermosphere can have temperatures of hundreds or thousands of degrees Celsius, the ISS does **not** experience ordinary atmospheric heating like an aircraft flying through dense air because there are extremely few molecules there.



5. EXOSPHERE

- The **exosphere** is the outermost region of Earth's atmosphere.
- The gases are extremely sparse, and there is no sharp boundary where the atmosphere suddenly ends. Instead, the exosphere **gradually becomes outer space**.
- Some satellites in **higher orbits** are located **well above the exosphere**



ATMOSPHERE

The atmosphere:

- **provides oxygen for respiration.**
- **provides carbon dioxide needed by plants for photosynthesis.**
- **protects Earth from harmful solar radiation.**
- **regulates Earth's temperature.**
- **supports the water cycle.**
- **allows weather and climate processes to occur and**
- **burns up many small meteoroids before they reach Earth's surface.**

AIR POLLUTION

Air pollution can come from **natural** and **human-made (anthropogenic)** sources.

A. NATURAL SOURCES

- Volcanic eruptions
- Forest fires caused by natural events
- Dust storms
- Pollen and plant materials
- Decomposition of organic matter
- Sea spray

B. HUMAN-MADE SOURCES

- Motor vehicles
- Factories and power plants
- Burning of coal, oil, and other fuels
- Open burning of waste
- Construction activities
- Agricultural activities
- Household fuel combustion.

POLLUTANTS

- Air pollutants can also be classified according to **how they are formed**.

Primary Pollutants

- **Primary pollutants** are released directly into the atmosphere from a source.

Pollutant	Common Source
Carbon monoxide (**CO**)	Incomplete combustion of fuels.
Sulfur dioxide (**SO_2**)	Burning sulfur-containing fuels.
Nitrogen oxides (**NO_x**)	Fuel combustion and vehicles.
Particulate matter (**PM**)	Combustion, construction, dust.
Volatile organic compounds (**VOC**s)	Fuel evaporation and industrial processes.